

```
import pandas as pd
```

```
df = pd.read_csv("Iris.csv")
```

```
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.shape
```

```
(150, 6)
```

```
df.columns
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
      'Species'],  
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 150 entries, 0 to 149  
Data columns (total 6 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0    Id              150 non-null    int64  
1    SepalLengthCm   150 non-null    float64  
2    SepalWidthCm    150 non-null    float64  
3    PetalLengthCm   150 non-null    float64  
4    PetalWidthCm    150 non-null    float64  
5    Species         150 non-null    object  
dtypes: float64(4), int64(1), object(1)  
memory usage: 7.2+ KB
```

```
df.tail()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	
count	150.000000	150.000000	150.000000	150.000000	150.000000	
mean	75.500000	5.843333	3.054000	3.758667	1.198667	
std	43.445368	0.828066	0.433594	1.764420	0.763161	
min	1.000000	4.300000	2.000000	1.000000	0.100000	
25%	38.250000	5.100000	2.800000	1.600000	0.300000	
50%	75.500000	5.800000	3.000000	4.350000	1.300000	
75%	112.750000	6.400000	3.300000	5.100000	1.800000	
max	150.000000	7.900000	4.400000	6.900000	2.500000	

```
df.isnull().sum()
```

```

      0
Id      0
SepalLengthCm  0
SepalWidthCm  0
PetalLengthCm  0
PetalWidthCm  0
Species      0

```

```
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['Species'].unique()
```

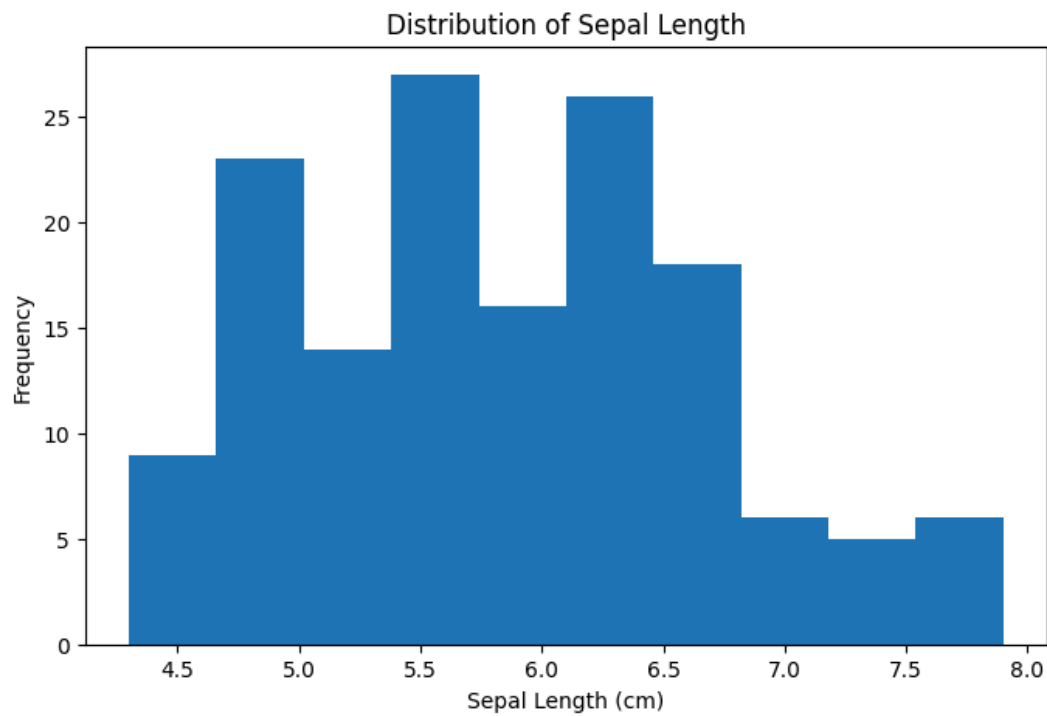
```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```

import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(df['SepalLengthCm'], bins=10)
plt.xlabel('Sepal Length (cm)')
plt.ylabel('Frequency')
plt.title('Distribution of Sepal Length')
plt.show()

```



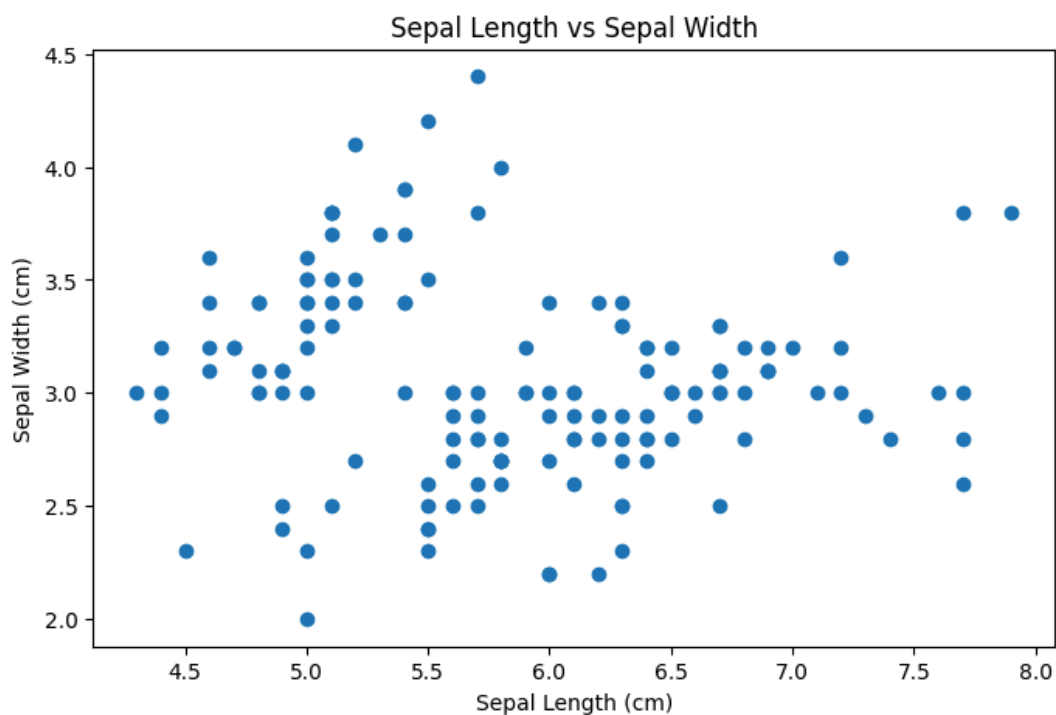
✓ Observation

The histogram shows that most flowers have a sepal length between approximately 5 and 6.5 cm. Very short and very long sepal lengths occur less frequently.

```
plt.figure(figsize=(8,5))

plt.scatter(df['SepalLengthCm'], df['SepalWidthCm'])

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Sepal Length vs Sepal Width')
plt.show()
```



Observation: The scatter plot shows that sepal length and sepal width do not have a strong linear relationship. The data points are spread across different combinations of sepal length and width.

```
plt.figure(figsize=(8,5))

df.boxplot(column='PetalLengthCm', by='Species')

plt.xlabel('Species')
plt.ylabel('Petal Length (cm)')
plt.title('Petal Length by Species')
plt.suptitle('')
plt.show()
```

<Figure size 800x500 with 0 Axes>

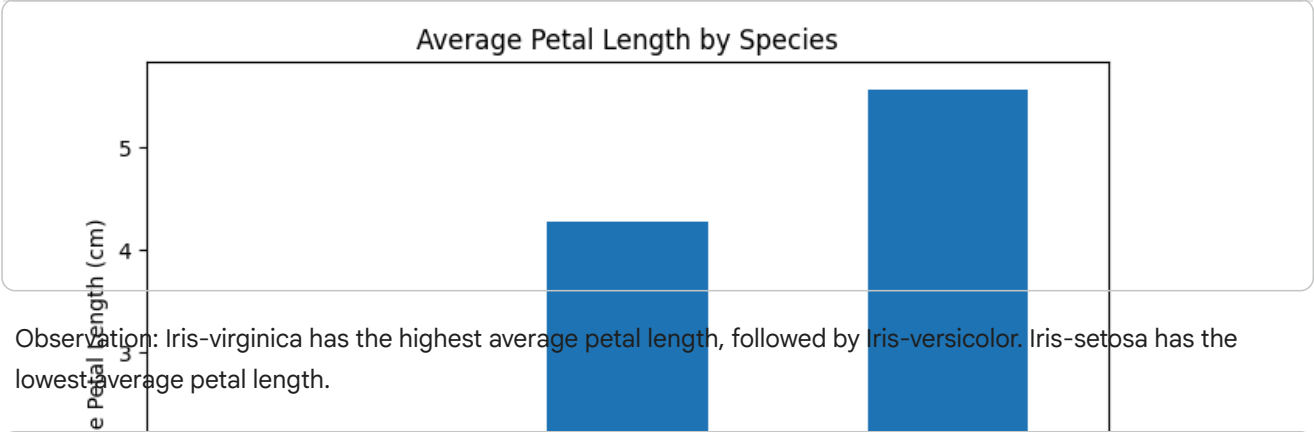


Observation: Iris-setosa has the smallest petal length, while Iris-virginica has the largest. Iris-versicolor falls between the other two species.

```
avg_petal_length = df.groupby('Species')['PetalLengthCm'].mean()

plt.figure(figsize=(8,5))
avg_petal_length.plot(kind='bar')

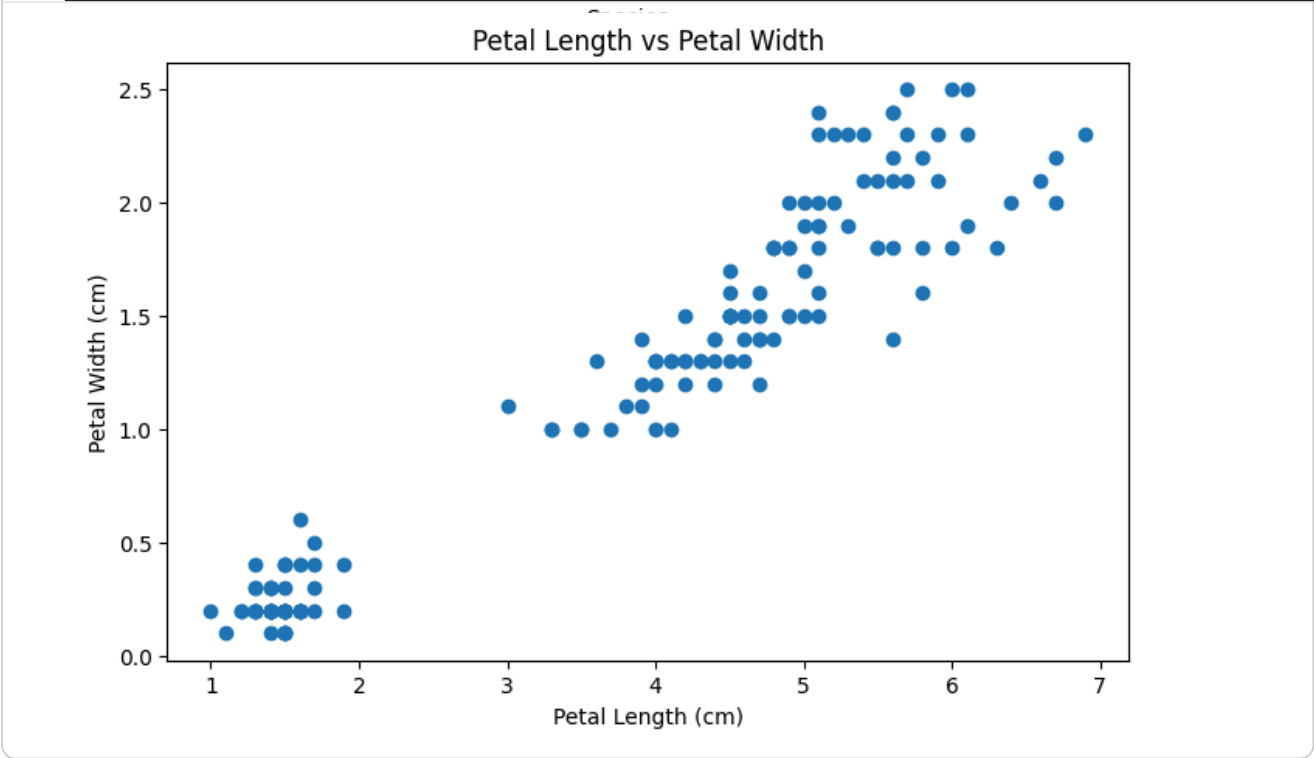
plt.xlabel('Species')
plt.ylabel('Average Petal Length (cm)')
plt.title('Average Petal Length by Species')
plt.xticks(rotation=0)
plt.show()
```



```
plt.figure(figsize=(8,5))

plt.scatter(df['PetalLengthCm'], df['PetalWidthCm'])

plt.xlabel('Petal Length (cm)')
plt.ylabel('Petal Width (cm)')
plt.title('Petal Length vs Petal Width')
plt.show()
```



Observation: Petal length and petal width show a strong positive relationship. Flowers with longer petals generally also have wider petals.

Start coding or [generate](#) with AI.

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****Top 3 Insights****

- 1. Petal measurements clearly distinguish the three Iris species. Iris-setosa generally has the smallest petals, while Iris-virginica has the largest.
- 2. Petal length and petal width have a strong positive relationship. As petal length increases, petal width generally increases as well.

Top 3 Insights

- 1. Petal measurements clearly distinguish the three Iris species. Iris-setosa generally has the smallest petals, while Iris-virginica has the largest.
- 2. Petal length and petal width have a strong positive relationship. As petal length increases, petal width generally increases as well.